```
import numpy as np



import matplotlib.pyplot as plt

import pandas as pd

import seaborn as sns



%matplotlib inline

dataset = pd.read_csv(r"C:\Users\srini\Downloads\11th - mlr\11th -
mlr\MLR\House_data.csv")

dataset.head()
Out[7]:
           id             date     price  ...     long  sqft_living15  sqft_lot15
0  7129300520  20141013T000000  221900.0  ... -122.257           1340        5650
1  6414100192  20141209T000000  538000.0  ... -122.319           1690        7639
2  5631500400  20150225T000000  180000.0  ... -122.233           2720        8062
3  2487200875  20141209T000000  604000.0  ... -122.393           1360        5000
4  1954400510  20150218T000000  510000.0  ... -122.045           1800        7503

[5 rows x 21 columns]



print(dataset.isnull().any())
id                 False
date               False
price              False
bedrooms           False
bathrooms          False
sqft_living        False
sqft_lot           False
floors             False
waterfront         False
view               False
condition          False
grade              False
sqft_above         False
sqft_basement      False
yr_built           False
yr_renovated       False
zipcode            False
lat                False
long               False
```

```
sqft_living15    False
sqft_lot15       False
dtype: bool

print(dataset.dtypes)
id                 int64
date              object
price            float64
bedrooms           int64
bathrooms        float64
sqft_living        int64
sqft_lot           int64
floors           float64
waterfront         int64
view               int64
condition          int64
grade              int64
sqft_above         int64
sqft_basement      int64
yr_built           int64
yr_renovated       int64
zipcode            int64
lat              float64
long             float64
sqft_living15      int64
sqft_lot15         int64
dtype: object

print(dataset.dtypes)
id                 int64
date              object
price            float64
bedrooms           int64
bathrooms        float64
sqft_living        int64
sqft_lot           int64
floors           float64
waterfront         int64
view               int64
condition          int64
grade              int64
sqft_above         int64
sqft_basement      int64
yr_built           int64
yr_renovated       int64
zipcode            int64
lat              float64
long             float64
sqft_living15      int64
sqft_lot15         int64
```

```
dtype: object

dataset = dataset.drop(['id','date'], axis = 1)

dataset = dataset.drop(['id','date'], axis = 1)
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[12], line 1
----> 1 dataset = dataset.drop(['id','date'], axis = 1)

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:5603, in
DataFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
   5455 def drop(
   5456     self,
   5457     labels: IndexLabel | None = None,
   (...)   5464     errors: IgnoreRaise = "raise",
   5465 ) -> DataFrame | None:
   5466     """
   5467     Drop specified labels from rows or columns.
   5468
   (...)   5601             weight  1.0     0.8
   5602     """
-> 5603     return super().drop(
   5604         labels=labels,
   5605         axis=axis,
   5606         index=index,
   5607         columns=columns,
   5608         level=level,
   5609         inplace=inplace,
   5610         errors=errors,
   5611     )

File ~\anaconda3\Lib\site-packages\pandas\core\generic.py:4810, in
NDFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
   4808 for axis, labels in axes.items():
   4809     if labels is not None:
-> 4810         obj = obj._drop_axis(labels, axis, level=level, errors=errors)
   4812 if inplace:
   4813     self._update_inplace(obj)

File ~\anaconda3\Lib\site-packages\pandas\core\generic.py:4852, in
NDFrame._drop_axis(self, labels, axis, level, errors, only_slice)
   4850         new_axis = axis.drop(labels, level=level, errors=errors)
   4851     else:
-> 4852         new_axis = axis.drop(labels, errors=errors)
   4853     indexer = axis.get_indexer(new_axis)
   4855 # Case for non-unique axis
   4856 else:

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:7136, in
```

```
Index.drop(self, labels, errors)
   7134 if mask.any():
   7135     if errors != "ignore":
-> 7136         raise KeyError(f"{labels[mask].tolist()} not found in axis")
   7137     indexer = indexer[~mask]
   7138 return self.delete(indexer)

KeyError: "['id', 'date'] not found in axis"

x=dataset.iloc[:,:-1]

y=dataset.iloc[:,4]

x=pd.get_dummies(x,dtype=int)

with sns.plotting_context("notebook",font_scale=2.5):
    g = 
sns.pairplot(dataset[['sqft_lot','sqft_above','price','sqft_living','bedrooms']],
                 hue='bedrooms', palette='tab20',size=6)
g.set(xticklabels=[]);
C:\Users\srini\anaconda3\Lib\site-packages\seaborn\axisgrid.py:2100: UserWarning: 
The `size` parameter has been renamed to `height`; please update your code.
  warnings.warn(msg, UserWarning)


Important
Figures are displayed in the Plots pane by default. To make them also appear inline 
in the console, you need to uncheck "Mute inline plotting" under the options menu of 
Plots.


with sns.plotting_context("notebook",font_scale=2.5):
    g = 
sns.pairplot(dataset[['sqft_lot','sqft_above','price','sqft_living','bedrooms']],
                 hue='bedrooms', palette='tab20',height=6)
g.set(xticklabels=[]);


X = dataset.iloc[:,1:].values
y = dataset.iloc[:,0].values

X = dataset.iloc[:,1:].values
y = dataset.iloc[:,0].values

from sklearn.cross_validation import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 1/3, 
random_state = 0)

---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
```

```
Cell In[20], line 1
----> 1 from sklearn.cross_validation import train_test_split
      2 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 1/3,
random_state = 0)

ModuleNotFoundError: No module named 'sklearn.cross_validation'

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 1/3,
random_state = 0)


from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train, y_train)

Out[22]: LinearRegression()

y_pred = regressor.predict(X_test)

import statsmodels.api as sm
X_opt = X[:, [0, 1, 2, 3, 4, 5,6,7,8,9,10,11,12,13,14,15,16,17]]

regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
regressor_OLS.summary()
Out[25]:
<class 'statsmodels.iolib.summary.Summary'>
"""
                                OLS Regression Results

==================================================================================
===
Dep. Variable:                      y   R-squared (uncentered):
0.905
Model:                            OLS   Adj. R-squared (uncentered):
0.905
Method:                 Least Squares   F-statistic:
1.211e+04
Date:                Mon, 11 May 2026   Prob (F-statistic):
0.00
Time:                        23:40:10   Log-Likelihood:
-2.9461e+05
No. Observations:               21613   AIC:
5.892e+05
Df Residuals:                   21596   BIC:
5.894e+05
Df Model:                          17

Covariance Type:            nonrobust
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
x1         -3.551e+04   1888.716    -18.802      0.000   -3.92e+04   -3.18e+04
x2          4.105e+04   3253.759     12.618      0.000    3.47e+04    4.74e+04
x3           110.2642      2.268     48.607      0.000     105.818     114.711
x4             0.1334      0.048      2.786      0.005       0.040       0.227
x5          5261.5471   3541.347      1.486      0.137   -1679.755    1.22e+04
x6          5.833e+05   1.74e+04     33.598      0.000    5.49e+05    6.17e+05
x7          5.236e+04   2128.298     24.600      0.000    4.82e+04    5.65e+04
x8          2.721e+04   2323.818     11.709      0.000    2.27e+04    3.18e+04
x9          9.548e+04   2145.492     44.503      0.000    9.13e+04    9.97e+04
x10           71.3928      2.238     31.902      0.000      67.006      75.779
x11           38.8714      2.624     14.813      0.000      33.728      44.015
x12        -2561.7953     68.006    -37.670      0.000   -2695.092   -2428.498
x13           20.4187      3.646      5.600      0.000      13.272      27.566
x14         -519.0756     17.826    -29.119      0.000    -554.016    -484.136
x15         6.022e+05   1.07e+04     56.106      0.000    5.81e+05    6.23e+05
x16        -2.179e+05   1.31e+04    -16.683      0.000   -2.44e+05   -1.92e+05
x17           23.0994      3.392      6.811      0.000      16.452      29.747
x18           -0.3761      0.073     -5.137      0.000      -0.520      -0.233
==============================================================================
Omnibus:                    18403.146   Durbin-Watson:                   1.991
Prob(Omnibus):                  0.000   Jarque-Bera (JB):          1873534.498
Skew:                           3.572   Prob(JB):                         0.00
Kurtosis:                      48.049   Cond. No.                     4.19e+17
==============================================================================

Notes:
[1] R² is computed without centering (uncentered) since the model does not contain a
constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[3] The smallest eigenvalue is 1.25e-21. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
"""

import statsmodels.api as sm
X_opt = X[:, [0, 1, 2, 3, 4,6,7,8,9,10,11,12,13,14,15,16,17]]
regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
regressor_OLS.summary()
Out[26]:
<class 'statsmodels.iolib.summary.Summary'>
"""
                                 OLS Regression Results                        

=================================================================================
===
Dep. Variable:                      y   R-squared (uncentered):                   
0.900
```

```
Model:                            OLS   Adj. R-squared (uncentered):
0.900
Method:                 Least Squares   F-statistic:
1.217e+04
Date:                Mon, 11 May 2026   Prob (F-statistic):
0.00
Time:                        23:41:39   Log-Likelihood:
-2.9516e+05
No. Observations:               21613   AIC:
5.903e+05
Df Residuals:                   21597   BIC:
5.905e+05
Df Model:                          16

Covariance Type:            nonrobust

==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
x1         -3.833e+04   1935.490    -19.804      0.000   -4.21e+04   -3.45e+04
x2          4.051e+04   3337.593     12.138      0.000     3.4e+04    4.71e+04
x3           112.2505      2.326     48.255      0.000     107.691     116.810
x4             0.1063      0.049      2.165      0.030       0.010       0.203
x5          5493.8277   3632.629      1.512      0.130   -1626.393    1.26e+04
x6          7.924e+04   2022.997     39.169      0.000    7.53e+04    8.32e+04
x7          2.813e+04   2383.557     11.801      0.000    2.35e+04    3.28e+04
x8          9.367e+04   2200.107     42.577      0.000    8.94e+04     9.8e+04
x9            75.1352      2.293     32.771      0.000      70.641      79.629
x10           37.1153      2.691     13.791      0.000      31.840      42.390
x11        -2524.4770     69.750    -36.193      0.000   -2661.192   -2387.762
x12           27.3206      3.734      7.316      0.000      20.001      34.640
x13         -528.8621     18.283    -28.927      0.000    -564.698    -493.026
x14         5.991e+05    1.1e+04     54.420      0.000    5.78e+05    6.21e+05
x15        -2.265e+05   1.34e+04    -16.906      0.000   -2.53e+05      -2e+05
x16           19.9451      3.478      5.735      0.000      13.129      26.762
x17           -0.3469      0.075     -4.620      0.000      -0.494      -0.200
==============================================================================
Omnibus:                    19193.801   Durbin-Watson:                   1.994
Prob(Omnibus):                  0.000   Jarque-Bera (JB):          2019493.632
Skew:                           3.827   Prob(JB):                         0.00
Kurtosis:                      49.733   Cond. No.                     4.46e+17
==============================================================================

Notes:
[1] R² is computed without centering (uncentered) since the model does not contain a
constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[3] The smallest eigenvalue is 1.1e-21. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

```
"""

import statsmodels.api as sm
X_opt = X[:, [0, 1, 2, 3,6,7,8,9,10,11,12,13,14,15,16,17]]
regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
regressor_OLS.summary()
Out[27]:
<class 'statsmodels.iolib.summary.Summary'>
"""
                                 OLS Regression Results                                 
=======================================================================================
===
Dep. Variable:                      y   R-squared (uncentered):                   
0.900
Model:                            OLS   Adj. R-squared (uncentered):              
0.900
Method:                 Least Squares   F-statistic:                          
1.298e+04
Date:                Mon, 11 May 2026   Prob (F-statistic):                       
0.00
Time:                        23:42:54   Log-Likelihood:                   
-2.9516e+05
No. Observations:               21613   AIC:                              
5.903e+05
Df Residuals:                   21598   BIC:                              
5.905e+05
Df Model:                          15                                         

Covariance Type:            nonrobust                                         

==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
x1         -3.844e+04   1934.126    -19.876      0.000   -4.22e+04   -3.47e+04
x2          4.186e+04   3217.290     13.010      0.000    3.55e+04    4.82e+04
x3           111.8761      2.313     48.368      0.000     107.342     116.410
x4             0.1041      0.049      2.121      0.034       0.008       0.200
x5          7.938e+04   2020.785     39.283      0.000    7.54e+04    8.33e+04
x6          2.782e+04   2374.888     11.714      0.000    2.32e+04    3.25e+04
x7          9.402e+04   2188.126     42.969      0.000    8.97e+04    9.83e+04
x8            76.3846      2.139     35.714      0.000      72.192      80.577
x9            35.4915      2.468     14.381      0.000      30.654      40.329
x10        -2506.5676     68.739    -36.465      0.000   -2641.302   -2371.834
x11           27.5536      3.731      7.384      0.000      20.240      34.867
x12         -531.6192     18.192    -29.222      0.000    -567.278    -495.961
x13         6.006e+05    1.1e+04     54.765      0.000    5.79e+05    6.22e+05
x14        -2.279e+05   1.34e+04    -17.049      0.000   -2.54e+05   -2.02e+05
x15           19.1697      3.440      5.573      0.000      12.427      25.912
x16           -0.3518      0.075     -4.688      0.000      -0.499      -0.205
```

```
==============================================================================
Omnibus:                    19151.207   Durbin-Watson:                   1.994
Prob(Omnibus):                  0.000   Jarque-Bera (JB):          2002585.298
Skew:                           3.814   Prob(JB):                         0.00
Kurtosis:                      49.536   Cond. No.                     3.61e+17
==============================================================================

Notes:
[1] R² is computed without centering (uncentered) since the model does not contain a
constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[3] The smallest eigenvalue is 1.68e-21. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
"""
```